10.2 The Stacked-Ball Effect

WORKSHEET

The question

(i) In an experiment, a particle A of mass m is at rest on a smooth horizontal table. A particle B of mass bm, where $b > 1$, is projected along the table directly towards A with speed u. The collision is perfectly elastic.

Find an expression for the speed of A after the collision in terms of b and u, and show that, irrespective of the relative masses of the particles, A cannot be made to move at twice the initial speed of B.

(ii) In a second experiment, a particle B_1 is projected along the table directly towards A with speed u. This time, particles $B_2, B_3, \ldots, B_n$ are at rest in order on the line between B_1 and A. The mass of B_i $(i = 1, 2, \ldots, n)$ is $\lambda^{n+1-i}m$, where $\lambda > 1$. All collisions are perfectly elastic. Show that, by choosing n sufficiently large, there is no upper limit on the speed at which A can be made to move.

In the case $\lambda = 4$, determine the least value of n for which A moves at more than $20u$. You may use the approximation $\log_{10} 2 \approx 0.30103$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself From Section 10.1: if a mass M moving at speed u hits a stationary mass m elastically, what is the speed of m afterwards?

STAGE 1 The speed limit

Ask yourself How does $\frac{2b}{b+1}$ compare with 2? What happens as $b \to \infty$?

STAGE 2 Passing it along a chain

Ask yourself In part (ii), each particle is λ times as heavy as the next one. What does part (i) say about each collision in the chain?

Ask yourself Is $\frac{2\lambda}{\lambda+1}$ bigger or smaller than 1?

Try this next

1 **The real thing.** (From STEP I, 2014.) A ball of mass M and radius R is released from rest with its centre at height $H + R$, with a ball of mass m and radius r on top. The coefficient of restitution for every impact is e. Show that, bouncing alone, the large ball would rise until its centre is at height $R + He^2$. Given $R = 0.2$, $r = 0.05$, $H = 1.8$, $e = \frac{2}{3}$ and that the small ball's centre reaches height 4.5, show that $M/m = 9$.

2 **An inelastic chain.** (From STEP I, 2017.) Particles $P_1, P_2, \ldots$ of masses $\lambda m, \lambda^2 m, \ldots$ lie at $x = 1, 2, \ldots$, and a particle of mass m hits P_1 at speed u; the coefficient of restitution is e. Show that P_1 moves off at $\frac{1+e}{1+\lambda}u$. Find the speed of P_n after its first collision, and show that if $e > \lambda$ each particle is involved in exactly two collisions. What happens if $\lambda e = 1$?

3 **Three balls.** Three perfectly elastic balls are stacked, each much lighter than the one below. Show that the top ball leaves at $7U$ and rises to $49H$. With k balls, show that the top one leaves at $(2^k - 1)U$.

4 **Where does the energy come from?** In part (ii), show that the total kinetic energy is conserved even though A's speed is unbounded. What happens to the heavy particles?

Further reading

- **David Morin, *Introduction to Classical Mechanics*** (Cambridge University Press), Chapter 5, which includes the stacked-ball problem.
- **W. R. Mellen, "Superball rebound projectiles"**, *American Journal of Physics* 36 (1968). Short and fun.
- **Gary Flandro, "Fast reconnaissance missions to the outer solar system utilizing energy derived from the gravitational field of Jupiter"**, *Astronautica Acta* 12 (1966). The paper that proposed the Voyager "grand tour".

The full guided solution, and what the question leads to at university, is in Section 10.2:
https://beyondstep.keithmcnulty.org/ch10/stacked-balls.html